

TODAY'S SESSION

Building a Single-Page Application

with HTML · CSS · JavaScript

Student Guide

Get Techy With Lucky | Instructor: Lucky Nakola

WHAT YOU WILL ACHIEVE TODAY

Understand what a Single-Page Application (SPA) is and why it matters.

Build a complete SPA from scratch -- no frameworks, no libraries, just clean code.

Apply HTML structure, CSS styling, and JavaScript navigation in one real project.

Walk away with working code you can run in any browser today.

1. What Is a Single-Page Application?

A Single-Page Application (SPA) is a website that loads ONE HTML file and then dynamically swaps content in and out as the user navigates -- without ever reloading the browser page.

1.1 Traditional Website vs SPA

Aspect	Traditional Site	SPA
How pages load	Each click loads a completely new HTML file from the server.	JavaScript swaps content inside the same page -- no server reload.
Speed feel	Small delay on every page change (server round-trip).	Instant -- feels like a native app.
Files needed	One .html file per page.	One index.html + JavaScript controls everything.
Examples	Simple brochure sites, blogs.	Gmail, Twitter, portfolio SPAs, dashboards.

1.2 How Does JavaScript Make It Work?

The trick is simple: all your 'pages' are actually hidden `<div>` sections sitting inside one HTML file. JavaScript listens for navigation clicks, hides the current section, and shows the requested one -- all in milliseconds.

THE CORE IDEA

1. One index.html file holds ALL sections (home, about, services, contact).
2. CSS hides all sections by default: `display: none;`
3. JavaScript shows only the active section: `display: block;`
4. Navigation links call a JavaScript function to switch sections.

1.3 Why Learn SPAs First?

- You build with skills you already have -- HTML, CSS, JS. No extra tools needed.
- You understand the concept that powers React, Vue, and Angular frameworks.
- A polished SPA portfolio site impresses clients and employers immediately.
- It is fast, mobile-friendly, and runs from a single file -- easy to host anywhere.

KEY TERM

Routing -- the system that decides which section to show based on a navigation action.

In a real framework, routing is complex. Today, we build our own simple router in about 10 lines of JavaScript.

2. The HTML Structure

Every SPA starts with a single index.html file. The key insight is that you write ALL your page sections inside this one file, wrapped in <section> tags with unique IDs. Only one section is visible at a time.

2.1 The index.html Skeleton

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Portfolio SPA</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <!-- NAVIGATION (always visible) -->
  <header>
    <div class="logo">Lucky Nakola</div>
    <nav>
      <a href="#" onclick="showPage('home')">Home</a>
      <a href="#" onclick="showPage('about')">About</a>
      <a href="#" onclick="showPage('services')">Services</a>
      <a href="#" onclick="showPage('contact')">Contact</a>
    </nav>
    <div class="hamburger" onclick="toggleMenu()">&#9776;</div>
  </header>

  <!-- ALL PAGES LIVE HERE -->
  <main>
    <section id="home" class="page active"><!-- home content --></section>
    <section id="about" class="page"><!-- about content --></section>
    <section id="services" class="page"><!-- services content --></section>
    <section id="contact" class="page"><!-- contact content --></section>
  </main>

  <footer>
    <p>&copy; 2025 Lucky Nakola. All rights reserved.</p>
  </footer>

  <script src="app.js"></script>
</body>
</html>
```

2.2 Anatomy of a Section

Each `<section>` tag becomes one 'page'. Notice:

- `id="home"` -- the unique name JavaScript will use to find it.
- `class="page"` -- all sections share this class so CSS can hide them together.
- `class="page active"` -- only ONE section gets the 'active' class at a time.

INSTRUCTOR TIP

Write out the full content for each section (hero, about text, service cards, contact form) inside the appropriate `<section>` tags. The structure above is your starting skeleton -- flesh it out with real content as you code.

3. The CSS Styling

The CSS does two critical jobs: (1) it hides all sections by default, and (2) it shows only the active section. Everything else is normal styling for a professional-looking site.

3.1 The SPA Switch Logic -- The Most Important CSS

```
/* -- SPA ROUTING LOGIC ----- */
.page {
  display: none;          /* ALL sections hidden by default */
}

.page.active {
  display: block;          /* Only the active section is visible */
  animation: fadeIn 0.4s ease; /* Smooth fade on page switch */
}

@keyframes fadeIn {
  from { opacity: 0; transform: translateY(12px); }
  to   { opacity: 1; transform: translateY(0);   }
}
```

3.2 Header & Navigation CSS

```
/* -- RESET & BASE ----- */
* { margin: 0; padding: 0; box-sizing: border-box; }
body { font-family: 'Arial', sans-serif; color: #1f2937; }

/* -- STICKY HEADER ----- */
header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 16px 40px;
  background: #1b3a6b;
  position: sticky;
  top: 0;
  z-index: 100;
}

.logo { color: #fff; font-size: 1.4rem; font-weight: 700; }

nav a {
  color: #fff;
```

```
text-decoration: none;
margin-left: 28px;
font-size: 0.95rem;
transition: color 0.2s;
}
nav a:hover { color: #93c5fd; }
nav a.active { color: #fbbf24; font-weight: 700; border-bottom: 2px solid #fbbf24;
}
```

3.3 Responsive Hamburger Menu

```
.hamburger { display: none; font-size: 1.6rem; color: #fff; cursor: pointer; }

@media (max-width: 768px) {
  .hamburger { display: block; }
  nav { display: none; flex-direction: column; position: absolute;
    top: 60px; right: 0; background: #1b3a6b; width: 200px;
    padding: 16px; }
  nav.open { display: flex; }
  nav a { margin: 8px 0; }
}
```

3.4 Hero Section CSS

```
.hero {
  min-height: 90vh;
  display: flex;
  flex-direction: column;
  justify-content: center;
  align-items: center;
  text-align: center;
  background: linear-gradient(135deg, #1b3a6b 0%, #0d9488 100%);
  padding: 60px 24px;
}
.hero h1 { font-size: clamp(2rem, 5vw, 3.5rem); color: #fff; margin-bottom: 16px; }
.hero p { font-size: 1.15rem; color: #cbd5e1; max-width: 540px; margin-bottom: 32px; }
.btn {
  padding: 14px 36px; background: #fbbf24; color: #1b3a6b;
  border-radius: 6px; font-weight: 700; font-size: 1rem;
  text-decoration: none; transition: background 0.2s;
}
.btn:hover { background: #f59e0b; }
```

4. The JavaScript Router

This is the brain of the SPA. The router is a JavaScript function that hides all sections, then shows the one that was clicked. It also highlights the correct nav link and saves your position so the browser Back button can work.

4.1 The Core Router Function

```
// app.js

// -- SHOW A PAGE -----
function showPage(pageId) {

  // 1. Hide ALL sections
  document.querySelectorAll('.page').forEach(section => {
    section.classList.remove('active');
  });

  // 2. Show the requested section
  document.getElementById(pageId).classList.add('active');

  // 3. Highlight the correct nav link
  document.querySelectorAll('nav a').forEach(link => {
    link.classList.remove('active');
  });
  document.querySelector(`nav a[onclick*='${pageId}']`).classList.add('active');

  // 4. Scroll back to top
  window.scrollTo({ top: 0, behavior: 'smooth' });
}

// -- LOAD HOME ON START -----
document.addEventListener('DOMContentLoaded', () => {
  showPage('home');
});
```

4.2 Mobile Hamburger Toggle

```
// -- TOGGLE MOBILE MENU -----
function toggleMenu() {
  document.querySelector('nav').classList.toggle('open');
}

// Close menu when a nav link is clicked on mobile
```



```
document.querySelectorAll('nav a').forEach(link => {
  link.addEventListener('click', () => {
    document.querySelector('nav').classList.remove('open');
  });
});
```

4.3 Contact Form Handler

```
// -- CONTACT FORM VALIDATION -----
document.getElementById('contactForm').addEventListener('submit', function(e) {
  e.preventDefault(); // Stop page reload

  const name    = document.getElementById('name').value.trim();
  const email    = document.getElementById('email').value.trim();
  const message = document.getElementById('message').value.trim();

  if (!name || !email || !message) {
    document.getElementById('formMsg').textContent =
      'Please fill in all fields.';
    document.getElementById('formMsg').style.color = 'red';
    return;
  }

  // Success
  document.getElementById('formMsg').textContent =
    'Thank you, ' + name + '! We will be in touch.';
  document.getElementById('formMsg').style.color = 'green';
  this.reset();
});
```

HOW THE ROUTER WORKS -- STEP BY STEP

- Step 1 User clicks 'About' in the navigation.
- Step 2 onclick="showPage('about')" calls our JavaScript function.
- Step 3 The function removes 'active' from ALL .page sections (hides everything).
- Step 4 The function adds 'active' to the #about section (shows it with a fade).
- Step 5 The correct nav link gets highlighted in amber.
- Step 6 Page scrolls smoothly to the top.

5. Files, Sections & Quick Reference

5.1 Your Project File Structure

```
my-spa-portfolio/  
|- index.html      <- The ONE HTML file (all sections live here)  
|- style.css       <- All your CSS styles  
|- app.js          <- Your JavaScript router + interactions
```

That's it -- 3 files. No build tools, no npm, no framework. Open index.html in a browser and your SPA works immediately.

5.2 What Goes Inside Each Section

Section ID	Suggested Content
#home	Hero with headline, subheadline, and CTA button. Optional: stats row (e.g. 3 Years Experience, 20+ Projects).
#about	Two-column layout: profile photo on one side, bio + skills list on the other.
#services	3-column card grid: each card has an icon, title, and short description.
#contact	Contact form (name, email, message, submit) + inline validation message div.

5.3 Essential CSS Variables to Set at the Top

```
/* style.css -- paste at the very top */  
:root {  
  --primary:    #1b3a6b;    /* Navy -- main brand colour */  
  --accent:     #0d9488;    /* Teal -- highlights, icons */  
  --cta:        #fbbf24;    /* Amber -- buttons */  
  --text:       #1f2937;    /* Dark grey -- body text */  
  --light:      #f3f4f6;    /* Off-white -- section backgrounds */  
  --radius:     8px;        /* Border radius for cards/buttons */  
  --shadow:     0 4px 16px rgba(0,0,0,0.08); /* Card shadow */  
}
```

5.4 Service Card Template (copy & repeat 3x)

```
<!-- Inside the #services section -->  
<div class="services-grid">  
  
  <div class="card">  
    <div class="icon">&#128187;</div>    <!-- emoji or SVG icon -->  
    <h3>Web Design</h3>
```

```
<p>Beautiful, responsive websites built with clean, modern HTML and CSS.</p>
</div>
```

```
<div class="card">
  <div class="icon">&#128200;</div>
  <h3>Frontend Dev</h3>
  <p>Interactive SPAs and dynamic interfaces using Vanilla JavaScript.</p>
</div>
```

```
<div class="card">
  <div class="icon">&#128241;</div>
  <h3>Mobile-First</h3>
  <p>Every site I build works perfectly on phones, tablets, and desktops.</p>
</div>
```

```
</div>
```

```
/* Card grid CSS */
.services-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(240px, 1fr));
  gap: 24px;
  padding: 60px 40px;
}
.card {
  background: #fff;
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  padding: 32px 24px;
  text-align: center;
  transition: transform 0.2s;
}
.card:hover { transform: translateY(-6px); }
.icon { font-size: 2.4rem; margin-bottom: 12px; }
```

6. Knowledge Check -- Quiz

Answer the following 12 questions on your own before discussing with a classmate. Write your answers in the space provided or on a separate sheet.

INSTRUCTIONS

Answer all 12 questions. Some are multiple choice, some are short answer, and some ask you to write a small piece of code.

Section A -- Understanding (Questions 1-5)

1. What does SPA stand for, and in one sentence, explain what makes it different from a traditional multi-page website?
2. In an SPA, where does ALL the page content live?
 - a) In separate .html files on the server
 - b) In one index.html file as hidden sections
 - c) In a database that PHP fetches on demand
 - d) In the browser's local storage
3. Which TWO CSS classes work together to control which section is visible? Explain what each one does.
4. What CSS property do we use to hide all .page sections by default? What value is used to show the active one?
5. Why is position: sticky; used on the header? What problem does it solve?

Section B -- Code Reading (Questions 6-9)

6. Look at this JavaScript function. What does it do? Explain each of the 4 steps in plain English:

```
function showPage(pageId) {  
  document.querySelectorAll('.page').forEach(s => s.classList.remove('active'));  
  document.getElementById(pageId).classList.add('active');  
  document.querySelectorAll('nav a').forEach(a => a.classList.remove('active'));  
  document.querySelector(`nav a[onclick*='${pageId}']`).classList.add('active');  
}
```

7. What does e.preventDefault() do inside the form submit event listener? What would happen if you left it out?

8. What is the purpose of the `@keyframes fadeIn` animation in the CSS? Rewrite it in plain English.
9. What does `document.querySelector('nav').classList.toggle('open')` do? When is it called?

Section C -- Application (Questions 10-12)

10. Write the HTML for a navigation link that, when clicked, shows the section with `id="portfolio"`.
11. Write the CSS media query that hides the nav and shows the hamburger icon on screens 768px wide or narrower.
12. A classmate says: 'I will just use a different .html file for each page -- it's easier.' Give TWO reasons why building an SPA instead is a better choice for a personal portfolio.

7. Today's Build Project

Build Your Own SPA Portfolio Website

Using only index.html, style.css, and app.js -- build a complete, working personal portfolio SPA today. This is your deliverable for the session.

Project Requirements

HTML -- Structure

1. One index.html file containing ALL four sections: home, about, services, contact.
2. A sticky header with a logo (your name) and navigation links for all four sections.
3. A footer with your name and copyright year.
4. A contact form with name, email, and message fields plus a submit button.
5. A hamburger icon in the header (visible on mobile only).

CSS -- Styling

6. An external style.css file linked in the <head>.
7. SPA routing logic: .page { display: none } and .page.active { display: block }.
8. A fadeIn animation on page transitions.
9. Hero section with a gradient background, large headline, and a CTA button.
10. A 3-card services grid using CSS Grid (auto-fit, minmax).
11. Card hover effect (translateY lift + shadow).
12. Responsive design -- hamburger menu works on mobile screens.

JavaScript -- Behaviour

13. A showPage(pageId) function that switches sections and highlights the active nav link.
14. A toggleMenu() function for the mobile hamburger.
15. Contact form validation -- all three fields must be filled before submission.
16. A success or error message displayed beneath the form (no alert() popups).
17. Home page loads automatically when the site first opens.

Marking Rubric

Criterion	Marks	Awarded
HTML: Correct structure with 4 sections, header, footer, form	20	
CSS: SPA routing logic (hide/show + fade animation)	20	

Criterion	Marks	Awarded
CSS: Hero gradient, card grid, hover effects, responsive layout	20	
JS: showPage() router -- switches sections, highlights nav link	20	
JS: Form validation with on-page success/error message	10	
JS: Mobile hamburger menu toggle works	5	
General: Clean code, consistent indentation, readable	5	
TOTAL	100	

SUBMISSION CHECKLIST -- tick each before you hand in

- ☐ index.html, style.css, and app.js are all in the same folder.
- ☐ The site opens correctly by double-clicking index.html in a browser.
- ☐ All four nav links switch sections without a page reload.
- ☐ The active nav link is highlighted differently from the others.
- ☐ The contact form shows a message without reloading the page.
- ☐ The hamburger menu opens and closes on a narrow screen.
- ☐ No browser console errors (press F12 to check).

Good luck -- and enjoy the build!

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